

Ratio and Proportion — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give every ratio in its simplest form, and convert to a common unit before simplifying.
- The mark for each question is shown in square brackets on the right.
- Several questions do not give you the total. Work out what one part is worth first — everything else follows from it.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – WHEN THE TOTAL IS NOT GIVEN

1. Red and blue beads are threaded in the ratio 3 : 8. There are 96 blue beads.

(a) How many red beads are there?

[2]

(b) How many beads are there altogether?

[1]

2. Two numbers are in the ratio 5 : 9. The difference between them is 36. Find both numbers.

[3]

The difference is measured in parts too. How many parts is it?

- 3.** An amount of money is shared in the ratio 3 : 5 : 7. The largest share is ₹840. Find [3]
the total amount that was shared.

SECTION 2 – COMBINING TWO RATIOS

- 4.** For three quantities A, B and C you are told that $A : B = 2 : 3$ and $B : C = 4 : 5$.

(a) Write the ratio $A : B : C$ in its simplest form. [2]

(b) 105 marbles are shared in the ratio $A : B : C$. Find each share. [2]

B is 3 parts in one ratio and 4 in the other. Scale both until the two figures for B agree.

- 5.** In a school the ratio of teachers to students is 1 : 18, and the ratio of students to computers is 3 : 1. Find the ratio of teachers to computers, in its simplest form. [3]

SECTION 3 – MULTI-STEP

- 6.** 6 workers can build a wall in 10 days. All the workers work at the same rate.

(a) How long would 4 workers take to build the same wall? [2]

(b) How many workers would be needed to build it in 5 days? [1]

Check the direction of your answer to (a) before you write it down. Fewer workers cannot be faster.

7. A sum of money is shared between Anil and Bina in the ratio 5 : 3. Anil then gives ₹120 of his share to Bina, and they now have equal amounts. How much money was shared altogether? [4]

8. 4 pipes fill a tank in 9 hours. All the pipes deliver water at the same rate.

(a) How long would 6 pipes take to fill the tank? [2]

(b) A student writes “6 pipes, so $9 \times 6 \div 4 = 13.5$ hours.” Explain the mistake in one sentence. [1]

9. Rice and lentils are mixed in the ratio 5 : 2 by mass. A shop makes 21 kg of the mixture. Rice costs ₹60 per kg and lentils cost ₹95 per kg. Find the total cost of the ingredients in the 21 kg of mixture. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.